

Customer Churn Prediction & Segmentation for a Digital Bank

From Business Understanding to a Leakage-Safe, Budget-Aware Retention Pipeline

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One in five customers has already left — the goal is the recoverable 10%

20.37%

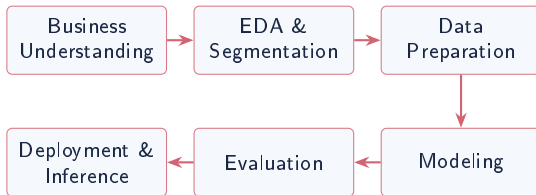
churn rate ($\approx 1:4$)

10,000

customers, 12 real columns

“Which customers are likely to churn next quarter, and in which segment — so retention is targeted, not sprayed?”

CRISP-DM pipeline followed end to end:



So what: Every downstream decision is train-only and leakage-audited — rigor is the product, not just the AUC.

Three business questions map to three analytical approaches

Objective: Frame the ask before modeling — avoid the top cause of wasted DS effort.

Technique: Question-to-technique mapping (Descriptive / Predictive / Prescriptive).

Question	Approach	Method
Q1 — <i>Who</i> is likely to churn?	Classification	Logistic baseline → tree ensembles → soft voting; budget-K operating point
Q2 — <i>Which segment</i> ?	Descriptive / Diagnostic	4 RFM-proxy segments + age_band×products heatmap + K-means cross-check
Q3 — <i>How</i> to target retention?	Prescriptive	Precision@K vs. retention budget; expected value; A/B test to prove causality

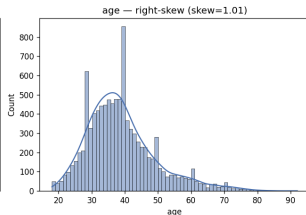
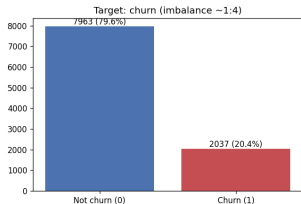
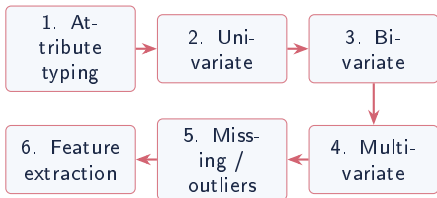
Governance guardrail — two-layer conclusion rule

Behavioral data is **synthetic** (generated *conditionally on* churn). It proves only that the **pipeline is correct**. All **business** conclusions are drawn from the 12 **real** columns only.

Clean data, one dominant driver: a six-pass read of 10,000 customers

Objective: Turn a raw table into a defensible “what drives churn” summary.

Technique: Six-step EDA: attributes → univariate → bivariate → multivariate → anomalies → features.



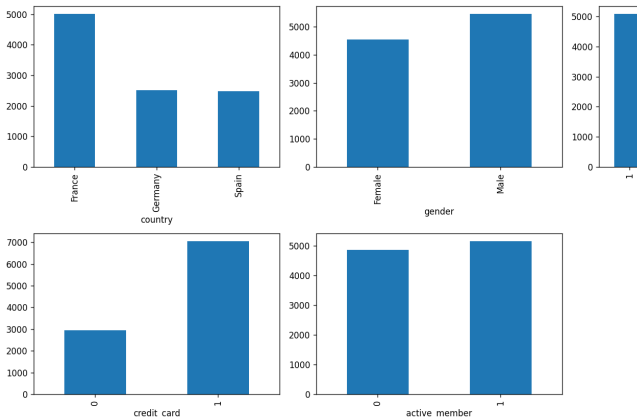
Insights & observations

- Quality is clean: **0 duplicates, 0 missing**; only age is right-skewed (≈ 1.0).
- Class imbalance confirmed: **20.37% churn** → carried to `class_weight`/SMOTE.
- **age** shows a **non-linear** churn relationship (peaks mid-life) — flagged for remediation.
- No numeric pair exceeds $|r| > 0.5$ → no worrying collinearity.

Product holdings — not demographics — dominate churn risk

Objective: Rank churn drivers on the real columns to focus modeling and retention.

Technique: Effect sizes: Cramér's V (categorical), η^2 (numeric); χ^2 /ANOVA significance.



Insights & observations

Effect-size ranking (real data):

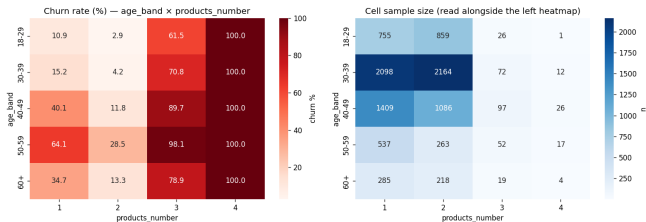
- **products_number $V=0.388 \gg$ rest**
- **country 0.174 > active_member 0.156 > gender 0.106**
- **age $\eta^2=0.081$ (non-linear)**
- **tenure, salary, credit_card ≈ 0 — not significant**

*The “new customers churn more” hypothesis is **not** supported ($\text{tenure } \eta^2 \approx 0$).*

Products ≥ 3 push churn toward 100%; risk peaks at ages 50–59

Objective: Expose the interaction the univariate view hides.

Technique: Churn-rate pivot heatmap over `age_band` $\times$ `products_number`.



Insights & observations

- `products=4` $\rightarrow$ **100%** churn; `products=3` $\rightarrow$ **82.7%** (small n : trust direction, wide CIs).
- Churn is **non-monotone** in age, peaking at **56%** for the 50–59 band.
- The two effects **compound**: inactive, multi-product, mid-life = the darkest cells.

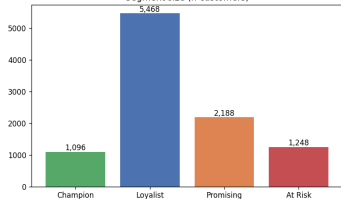
So what: These cells define the At-Risk list before any classifier runs.

Four rule-based segments split risk 13×: At Risk 70% vs. Champion 5.4%

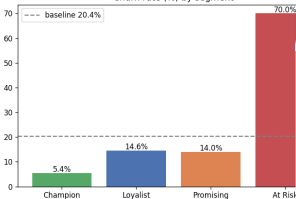
Objective: Answer Q2 — assign every customer to an actionable group on real columns.

Technique: Priority-ordered RFM-proxy rules (products, tenure, active, age); K-means cross-check.

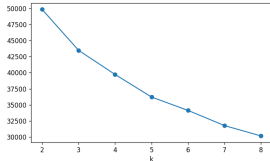
Segment size (n customers)



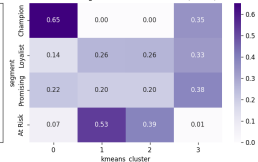
Churn rate (%) by segment



Elbow — inertia vs k



Rule-based segment × K-means cluster (row %)



Insights & observations

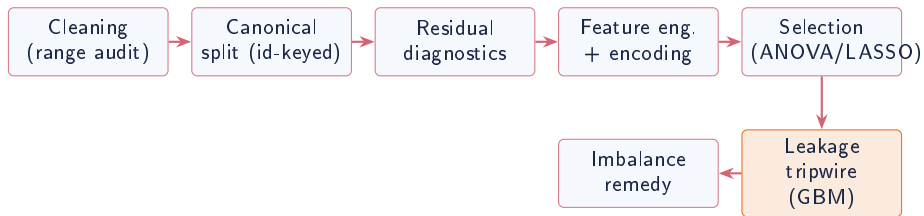
- **At Risk** — $n=1,248$, **70%** churn (products ≥ 3 or inactive 45+).
- **Champion** — $n=1,096$, **5.4%** churn (active, tenure ≥ 6 , 2 products).
- **Promising** (new, tenure ≤ 2) and **Loyalist** (rest) fill the middle.
- K-means (silhouette 0.25) recovers the same structure → rules aren't missing natural clusters.

So what: Each segment earns its own retention playbook

A leakage-safe pipeline: one canonical split, every decision train-only

Objective: Guarantee no test information contaminates any modeling decision.

Technique: Single `customer_id`-keyed split persisted to disk; assertions re-check it at every stage.



Cleaning result

0 duplicates, 0 out-of-range rows — the raw file was already clean. Cleaning is an **audited safety net**, not a fix.

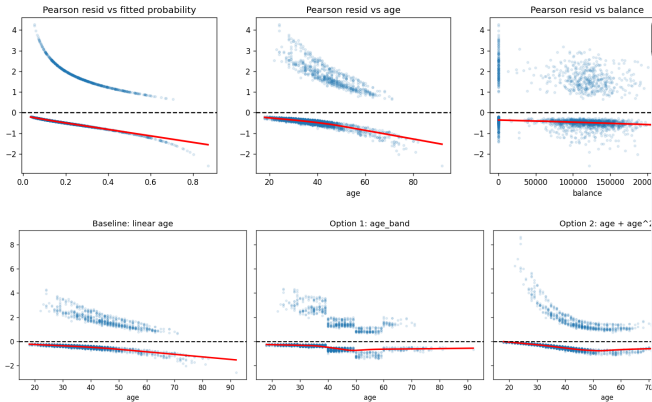
Split discipline (fix vs. v1)

v1 split twice with two mechanisms. v2 uses **one** split (`split_ids_v2.json`), reused everywhere with `assert`.

Churn is non-linear in age; age_band wins on AIC (7387→7063)

Objective: Diagnose and remediate the age non-linearity without leaking test data.

Technique: Pearson residuals from a logistic fit (train-only); age_band vs. age+age² compared by AIC.



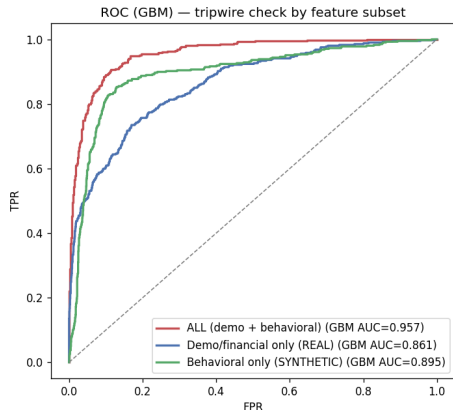
Insights & observations

- Binary target → OLS residplot is the **wrong** diagnostic; use Pearson $(y - \hat{p})/\sqrt{\hat{p}(1 - \hat{p})}$.
- AIC: linear 7387 → **age_band 7063** (winner) vs. age+age² 7082.
- Both flatten the LOWESS trend; age_band also communicates better (“risk peaks 50–59”).
- Decision rests on **8,000 train rows only** — v1’s leakage removed.

GBM leakage tripwire PASSES at $0.957 < 0.99$ — real-only signal is AUC 0.861

Objective: Prove the synthetic block cannot let a strong model “cheat”.

Technique: Multivariate check with GradientBoosting on ALL / demo-only / behavioral-only feature sets.



Insights & observations

- GBM (strongest class): max AUC-ROC $0.957 < 0.99 \rightarrow$ **PASS**.
- **Demo-only (real): AUC-ROC 0.861 / PR 0.704** — the only production-meaningful numbers.
- Behavioral-only 0.895 $\rightarrow$ pipeline proof; the ALL gain is from the synthetic block.

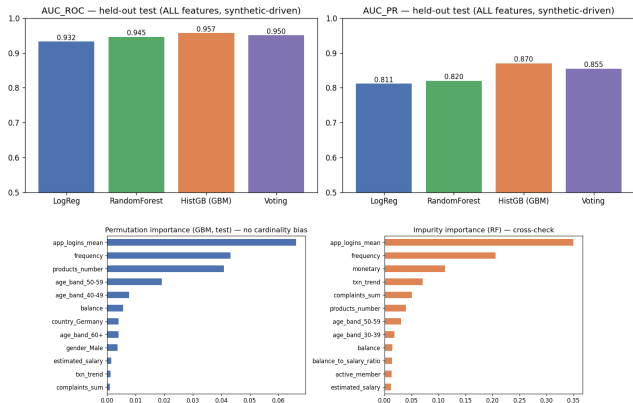
Imbalance decision

`class_weight` $\approx$ SMOTE (CV AUC-PR 0.804 vs. 0.802) $\rightarrow$ pick `class_weight`: simpler, no synthetic rows.

Soft-voting ensemble on real features: AUC-ROC 0.861 [0.840–0.881]

Objective: Select a champion that is both strong and defensible under imbalance.

Technique: Logistic → RandomForest / HistGBM → soft Voting; 5-fold CV; permutation vs. impurity importance.



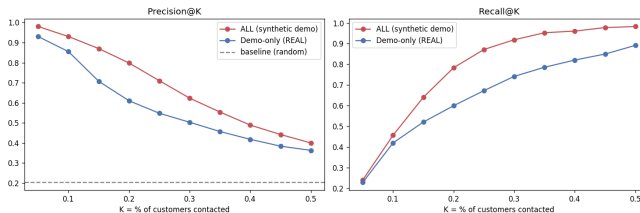
Insights & observations

- Champion chosen by **test AUC-PR** (right metric under 1:4 imbalance).
- Demo-only HistGBM: **AUC-ROC 0.861** [bootstrap 95% CI 0.840–0.881], **AUC-PR 0.704**.
- Permutation & impurity rankings **agree** on the top real drivers → **products_number**, **age_band**, **active_member**.
- ALL-feature “importances” are self-fulfilling (synthetic) — reported but not interpreted.

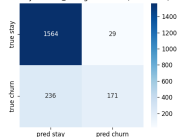
At a 10% outreach budget: $\text{Precision@10} = 0.855$, a $4.2\times$ lift

Objective: Turn a probability score into a retention decision under a fixed budget.

Technique: Precision@K / Recall@K vs. budget; calibration + Brier; bootstrap 95% CI (500 resamples).



mo-only model @ budget K=10% (thr=0.531)



Insights & observations

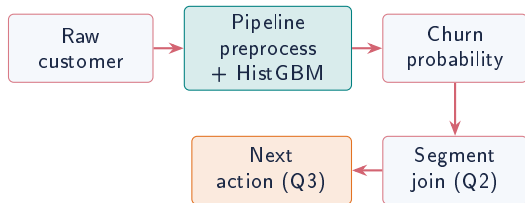
- AUC is not an operating decision — retention runs on a **budget of K%**.
- At K=10%: $\text{Precision@10} = 0.855$, lift $4.2\times$ over the 20.37% base rate.
- Threshold = 90th percentile of scores; revisit against the *actual* budget.
- Calibration checked (Brier) so probabilities can feed expected-value math.

So what: Contact the top decile → most churners at one-tenth the cost.

Deployed as one reproducible Pipeline with a model card

Objective: Package the real-feature model so others can reproduce or challenge it.

Technique: sklearn Pipeline (preprocess+model) + JSON model card (metrics, features, seed, caveats).



Model card (excerpt)

- HistGradientBoosting, **real features only** (20).
- seed=42, $n_{\text{train}}=8,000$, $n_{\text{test}}=2,000$.
- Test: AUC-ROC **0.861**, AUC-PR **0.704**; budget K=10%.
- Caveats: cross-sectional label; synthetic behavioral features **excluded**; uplift not causally validated.

Inference joins Q1 probability × Q2 segment into a Q3 action

Objective: Make the score operational — one row per customer, one recommended action.

Technique: Load saved pipeline → score test customers → join segment → map to a playbook.

Segment	Churn rate	Score	Recommended next action
At Risk	70%	High	Priority win-back: human outreach + tailored offer (A/B-tested)
Promising	mid	Med	Onboarding nudges to raise early activation
Loyalist	mid	Low–Med	Cross-sell to the 2-product “sweet spot”
Champion	5.4%	Low	Protect CLV: proactive perks, do <i>not</i> discount

From prediction to prescription

The system delivers a ranked top-K list *plus* the segment context that decides **which** retention lever to pull — not just a probability.

Conclusions are stratified: real-data insights vs. pipeline proof

Layer A — REAL data (business-safe)

- `products_number` ≥ 3 is the strongest red flag (4 $\rightarrow$ 100%, 3 $\rightarrow$ 82.7%) — a no-model rule.
- Churn peaks at **56%** for ages 50–59; `age_band` beats raw age (AIC 7387 $\rightarrow$ 7063).
- Segments: At Risk **70%** ($n=1,248$) vs. Champion **5.4%** ($n=1,096$).
- GBM demo-only: AUC-ROC **0.861**, Precision@10% **0.855**, lift **4.2 $\times$** .

Layer B — PIPELINE only (synthetic)

- Generator v3 keeps signal learnable (0.895) yet below the 0.99 tripwire even for GBM (max 0.957).
- Full chain runs end to end, reproducibly (seed 42, model card).
- “Top” behavioral features are self-fulfilling by construction — they prove the *code*, not the world.

So what: We publish externally *only* Layer A; Layer B documents that the machinery is trustworthy.

Open questions & next steps

Limitations

- Churn label is **cross-sectional** (no timestamp) — a 90-day window would relabel some customers.
- Behavioral evidence is **synthetic**; trend claims await real transaction history.
- Retention uplift is **assumed**, not measured.

Next steps

1. Swap synthetic tables for real transactions; redefine churn over 90 days; rerun from Data Prep §7.
2. **A/B test** the At-Risk offer to measure true uplift (replaces the 30% assumption).
3. Add `CalibratedClassifierCV` if probabilities feed expected value.
4. Monitor PSI/drift + AUC by monthly cohort in production.

So what: The only way to claim causal churn reduction is a live A/B test — everything above is observation & prediction.

Thank You

Questions & Discussion

Customer Churn Prediction & Segmentation for a Digital Bank — SOICT, HUST